Suraj Singh

PhD · ML Engineer & Data Scientist

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PROFILE

ML engineer and quantitative researcher with a PhD in computational physics and 3 years of applied machine learning experience in production environments. Background covers end-to-end model development from data ingestion through cloud deployment on GCP, with additional experience building and shipping React/TypeScript frontends. Comfortable working across the stack and communicating findings to both technical and non-technical stakeholders.

TECHNICAL SKILLS

Languages: Python (primary), SQL, TypeScript, MATLAB, Julia
ML / DL: PyTorch, LightGBM, XGBoost, scikit-learn, Transformers
GenAI / Agentic: LangChain, LangGraph, RAG, ChromaDB, NetworkX
Frontend: React, TypeScript, MSW (Mock Service Worker), Streamlit
Cloud / DevOps: GCP (Cloud Run, Cloud Scheduler, GCS), Docker, Cloudflare Pages, Wrangler, MLflow
Data: DuckDB, pandas, SQL, BigQuery
Other: Git/GitHub, FastAPI, Numerai era-based CV, Feature neutralisation

SELECTED PROJECTS

SE3 Electricity Price Forecast (live, production)

2024 – Present | Independent | github.com/surajsingh108/SE3-electricity-price-forecast

- End-to-end GCP pipeline: hourly data ingestion via Cloud Run Jobs from ENTSO-E and Open-Meteo, storage in DuckDB on GCS, weekly model retraining via Cloud Scheduler.
- LightGBM quantile model (p05/p50/p95) with 78 engineered features, Fourier harmonics, EWA price anchors, and V3 target neutralisation against a Ridge baseline. Zero data leakage by design.
- MLflow for experiment tracking and model versioning. Dockerised and deployed to Cloud Run. Live Streamlit dashboard at se3-dashboard-458800722354.europe-north1.run.app.

graph-rag-memory

2025 | Independent | github.com/surajsingh108/graph-rag-memory

- Graph-augmented RAG system with tiered source/derived memory in ChromaDB, decay-based compaction, and write-time deduplication using cosine similarity and residual novelty scoring.
- NetworkX knowledge graph enables multi-hop retrieval. Runs entirely on a local LLM with no external API dependency.

ATO Energy Platform Demo (React / Cloudflare)

2026 | Independent | ato-energy-demo.pages.dev

- Converted an existing React/TypeScript monorepo app into a zero-backend static demo using MSW 2.x to intercept all API calls with stateful synthetic data, covering IDSS audit trails, ML governance pipelines, EV impact dashboards, CGMES import/export, and line/transformer rating views.
- Deployed to Cloudflare Pages via Wrangler Direct Upload. Access-gated using Cloudflare Access one-time PIN email allowlist.
- Resolved same-origin API base issues, merged upstream feature branch, and maintained mock shape alignment across multiple deploy iterations.

Numerai Tournament (Financial ML)

2024 – Present | Ongoing

- Machine learning models for global equity market prediction using encrypted high-stakes financial datasets.
- Era-based cross-validation for drift detection; feature neutralisation for drift mitigation across market regimes.

PROFESSIONAL EXPERIENCE

ML Engineer (Contract)

Mar 2026 – Present | ATO Energy | Stockholm

- Physics-informed neural networks (PINNs) for Dynamic Line Rating: predicting real-time thermal behaviour of high-voltage power lines from sensor data.
- Digital twins of grid components used to validate ML predictions against simulated physical ground truth. Models used in live operational grid management decisions.

Postdoctoral Researcher

Oct 2023 – Oct 2025 | Stockholm University | Stockholm

- High-resolution numerical simulations and statistical analysis of fluid and atmospheric datasets. Managed collaborative workstreams with international partners.
- Synthesised technical findings into clear reports and presentations for non-specialist audiences; contributed to successful grant proposals.

Postdoctoral Researcher

Mar 2022 – Jun 2023 | University of Massachusetts Dartmouth | Massachusetts, USA

- Research on baroclinic instabilities and curved fronts. Applied advanced predictive modelling to generate new physical insights; co-authored successful government grant proposals.

EDUCATION

PhD & Masters, Aerospace Engineering | IIT Madras, India | 2021

B.Tech, Civil Engineering | NIT Trichy, India | 2013

PUBLICATIONS

S. Singh et al., "On baroclinic instability of curved fronts," *Dynamics of Atmospheres and Oceans*, 2025.

S. Singh and M. Mathur, "Diffusive effects in local instabilities of a baroclinic axisymmetric vortex," *Journal of Fluid Mechanics*, 2021.

S. Singh and M. Mathur, "Effects of Schmidt number on the short-wavelength instabilities in stratified vortices," *Journal of Fluid Mechanics*, 2019.